

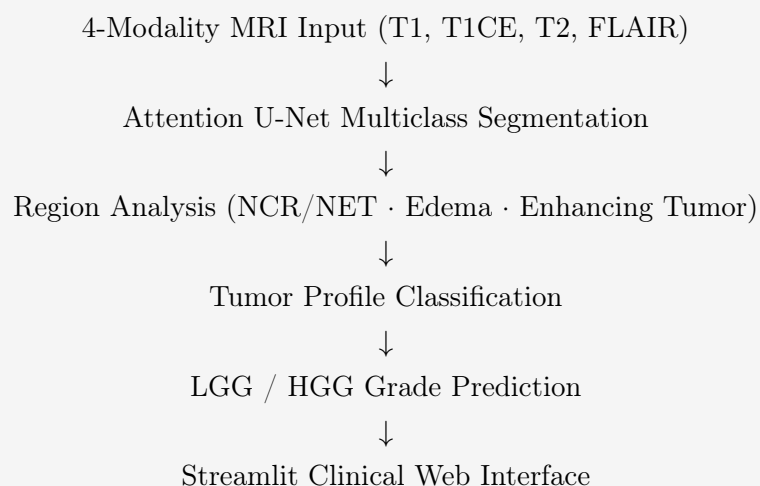
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# Automatic Brain Tumor Segmentation from MRI Images using U-Net

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*An End-to-End Deep Learning Pipeline for  
Multiclass Tumor Segmentation and Grade Classification*

## Pipeline Summary



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**Dataset:** BraTS 2021 (Brain Tumor Segmentation Challenge)

**Framework:** PyTorch, OpenCV, Streamlit

**Mean Dice:** 0.828 (competitive with 3D architectures)

**ET Dice:** 0.860 (after critical bug fix: 0.000  $\rightarrow$  0.860)

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# 1 Abstract

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Brain tumor detection and localization from Magnetic Resonance Imaging (MRI) scans is a critical task in clinical diagnosis and treatment planning. Manual segmentation by radiologists is time-consuming and subject to inter-observer variability. This project presents an automated deep learning framework for **multiclass brain tumor segmentation** and **grade classification** from multi-modal MRI scans.

An **Attention U-Net** architecture is trained to perform pixel-level segmentation of three tumor subregions — Necrotic Core/Non-Enhancing Tumor (NCR/NET), Peritumoral Edema, and Enhancing Tumor (ET) — achieving a Mean Dice of **0.828** and an ET Dice of **0.860**, competitive with published 3D architectures at significantly lower computational cost.

A critical data pipeline bug was discovered and resolved during development: naive alphabetical index slicing grouped four consecutive FLAIR slices per training sample instead of the four required modalities, causing T1CE to be absent from all training batches. This rendered Enhancing Tumor completely undetectable (Dice = 0.000). After implementing stem-based modality matching, ET Dice improved from **0.000 to 0.860** — an improvement of 0.860 without any architectural changes.

The complete system is deployed as a **Streamlit web application** providing real-time segmentation, tumor profile classification, and LGG/HGG grade prediction.

## 2 Introduction

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Brain tumors are abnormal cell growths in the brain that can significantly impair neurological function and patient survival. Gliomas, the most common primary brain tumors, are classified as Low Grade Glioma (LGG) or High Grade Glioma (HGG) based on aggressiveness and clinical prognosis.

Magnetic Resonance Imaging (MRI) provides high-resolution, multi-contrast anatomical information without ionizing radiation, making it the standard imaging modality for brain tumor diagnosis. However, manual segmentation of tumor regions requires significant specialist expertise and is time-consuming, with reported inter-observer variability of 15–28%.

### 2.1 Motivation

Current clinical workflows require:

- Manual delineation of tumor boundaries by neuroradiologists
- Multi-modal MRI interpretation across T1, T1CE, T2, and FLAIR sequences
- Subjective grading based on imaging features
- Invasive surgical biopsy for definitive tumor grading

This project proposes a fully automated, non-invasive pipeline that performs both tumor segmentation and grade classification directly from multi-modal MRI scans.

## 2.2 Key Contributions

### Research Contribution

1. A 2D multi-modal Attention U-Net achieving Mean Dice 0.828, competitive with 3D architectures at lower computational cost
2. Identification of a critical data pipeline failure mode where incorrect modality grouping renders Enhancing Tumor completely undetectable (Dice 0.000  $\rightarrow$  0.860 after fix)
3. An end-to-end pipeline from raw MRI to tumor grade prediction, deployed as a clinical web interface
4. Empirical evidence that attention gates provide negligible benefit (Mean Dice difference: 0.001) when multi-modal input provides sufficient discriminative signal

## 3 Dataset

### 3.1 BraTS 2021 Dataset

The **Brain Tumor Segmentation Challenge 2021 (BraTS 2021)** dataset is used for segmentation training and evaluation. It contains pre-operative multi-modal MRI scans of glioma patients with expert-annotated tumor subregion masks.

Table 1: BraTS 2021 Dataset Characteristics

| Property     | Value                               |
|--------------|-------------------------------------|
| Modalities   | T1, T1CE, T2, FLAIR (4 per patient) |
| Image format | PNG (converted from NIfTI .nii.gz)  |
| Resolution   | 240 × 240 pixels                    |
| Mask classes | 4 (Background, NCR/NET, Edema, ET)  |

### 3.2 MRI Modalities

Table 2: MRI Modality Descriptions and Clinical Significance

| Modality | Full Name                           | Key Clinical Use                                               |
|----------|-------------------------------------|----------------------------------------------------------------|
| T1       | T1-weighted                         | Anatomical reference, tissue contrast                          |
| T1CE     | Contrast-Enhanced T1                | <b>Enhancing Tumor detection</b> — bright contrast enhancement |
| T2       | T2-weighted                         | Fluid/water content, edema visualization                       |
| FLAIR    | Fluid Attenuated Inversion Recovery | Peritumoral edema, suppresses CSF signal                       |

### 3.3 Mask Label Definitions

Table 3: Segmentation Mask Class Labels

| Pixel Value | Class           | Description                         |
|-------------|-----------------|-------------------------------------|
| 0           | Background      | Healthy brain tissue and background |
| 1           | NCR/NET         | Necrotic core / non-enhancing tumor |
| 2           | Edema           | Peritumoral edema                   |
| 3           | Enhancing Tumor | Active tumor (bright on T1CE)       |

*Note: Raw BraTS masks use label 4 for Enhancing Tumor. This is remapped to 3 during preprocessing.*

### 3.4 LGG/HGG Dataset

For grade classification, the **BraTS 2021** dataset in HDF5 format is used, providing LGG/HGG labels per patient volume.

Table 4: Grade Classification Dataset Split

| Split        | LGG (0)       | HGG (1)       | Total         |
|--------------|---------------|---------------|---------------|
| Train        | 26,218        | 34,558        | 60,776        |
| Val          | 6,555         | 8,156         | 14,711        |
| <b>Total</b> | <b>32,773</b> | <b>42,714</b> | <b>75,487</b> |

## 4 Methodology

### 4.1 Pipeline Overview

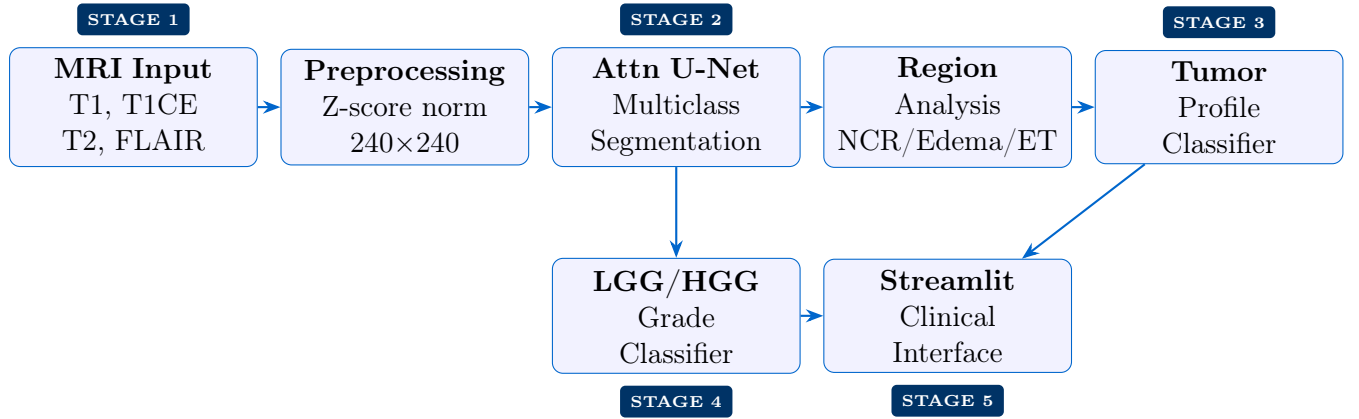


Figure 1: End-to-end brain tumor analysis pipeline

### 4.2 Data Preprocessing

1. **Image resizing:** All images resized to  $240 \times 240$  pixels
2. **Z-score normalization:** Applied per-channel on brain-masked pixels
3. **Brain masking:** Pixels with intensity  $> 0.1$  (after  $/255$  scaling) are considered brain tissue
4. **Mask remapping:** BraTS label 4 remapped to 3 for Enhancing Tumor
5. **Channel stacking:** Fixed order — T1, T1CE, T2, FLAIR (channels 0–3)

The z-score normalization formula applied to each channel  $c$  is:

$$\hat{x}_c = \frac{x_c - \mu_{\text{brain},c}}{\sigma_{\text{brain},c} + \epsilon} \quad (1)$$

where  $\mu_{\text{brain},c}$  and  $\sigma_{\text{brain},c}$  are computed only over brain-masked pixels, and  $\epsilon = 10^{-8}$  prevents division by zero. This normalization is critical for Enhancing Tumor detection — ET in T1CE appears as +2 to +4 standard deviations above the mean, providing a strong discriminative signal.

### 4.3 Data Augmentation

Table 5: Data Augmentation Strategy

| Augmentation        | Probability | Parameters                                                 |
|---------------------|-------------|------------------------------------------------------------|
| Horizontal Flip     | 0.5         | —                                                          |
| Vertical Flip       | 0.2         | —                                                          |
| Random Rotate 90°   | 0.5         | —                                                          |
| Brightness/Contrast | 0.3         | $\pm 0.2$ range                                            |
| Gaussian Noise      | 0.2         | —                                                          |
| Affine Transform    | 0.3         | translate $\pm 5\%$ , scale 0.9–1.1, rotate $\pm 15^\circ$ |

*Spatial augmentations (flip, rotate, affine) are applied identically across all 4 channels and the mask to preserve spatial alignment.*

### 4.4 Model Architecture

#### 4.4.1 Primary: Attention U-Net (Multiclass Segmentation)

The Attention U-Net [2] extends the standard U-Net by incorporating **attention gates** in the skip connections, allowing the network to focus on relevant spatial regions while suppressing background activations.

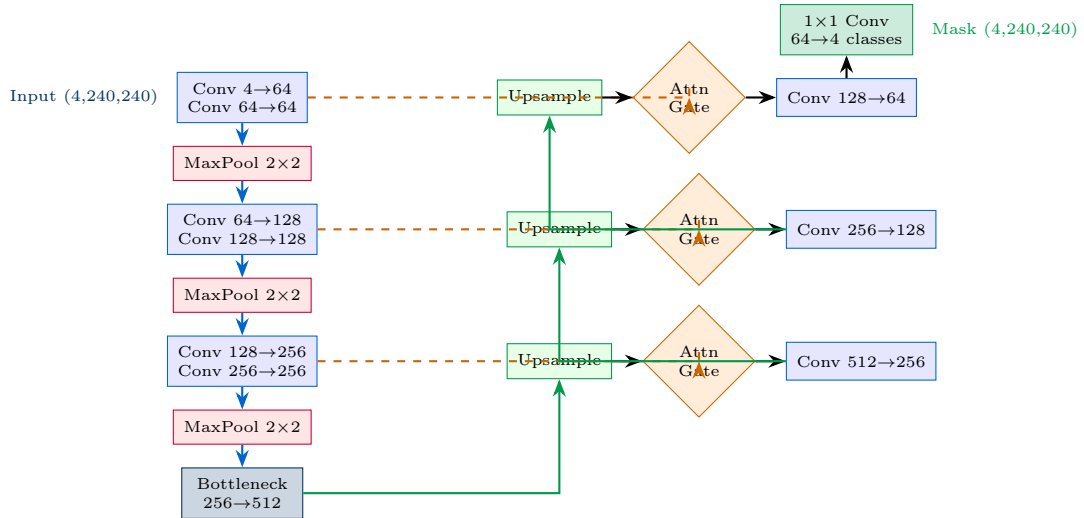


Figure 2: Attention U-Net architecture. Dashed arrows indicate skip connections with attention gating. Input is a 4-channel tensor (T1, T1CE, T2, FLAIR). Output is a 4-class probability map.

**Input:**  $(B, 4, 240, 240)$  — batch of 4-channel multi-modal tensors

**Output:**  $(B, 4, 240, 240)$  — per-pixel class probabilities, argmax gives final mask

#### 4.4.2 Tumor Profile Classifier

A CNN-based classifier predicts the overall tumor severity from a single T1CE slice:

Table 6: Tumor Severity Profile Classes

| Class | Label        | Description                                 |
|-------|--------------|---------------------------------------------|
| 0     | No Tumor     | No tumor regions detected in slice          |
| 1     | Edema Only   | Peritumoral edema present, no tumor core    |
| 2     | Core Present | Tumor core (NCR/NET $\pm$ edema), no ET     |
| 3     | Full Tumor   | All three regions including enhancing tumor |

#### 4.4.3 LGG/HGG Grade Classifier

A ResNet-18 [3] fine-tuned for binary classification:

- **Input:** T1CE grayscale slice (1, 240, 240) — first conv layer adapted from 3-channel to 1-channel
- **Output:** Binary prediction — LGG (0) or HGG (1)

- **Training:** Weighted Cross-Entropy to handle class imbalance (43% LGG vs 57% HGG)

## 4.5 Training Configuration

Table 7: Training Hyperparameters

| Parameter             | Value                              |
|-----------------------|------------------------------------|
| Input Resolution      | $240 \times 240$                   |
| Input Channels        | 4 (T1, T1CE, T2, FLAIR)            |
| Output Classes        | 4 (Background, NCR/NET, Edema, ET) |
| Batch Size            | 16                                 |
| Epochs (Segmentation) | 75                                 |
| Epochs (Classifier)   | 30                                 |
| Optimizer             | Adam                               |
| Learning Rate         | $1 \times 10^{-4}$                 |
| LR Scheduler          | StepLR (step=10, $\gamma=0.5$ )    |
| Loss Function         | Dice Loss + Cross-Entropy Loss     |
| Z-score Threshold     | Brain pixels $> 0.1$               |

## 4.6 Evaluation Metrics

**Dice Coefficient** (primary metric):

$$\text{Dice} = \frac{2 \cdot |P \cap G|}{|P| + |G|} \quad (2)$$

**Intersection over Union (IoU / Jaccard):**

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} \quad (3)$$

where  $P$  is the predicted region and  $G$  is the ground truth region.

## 5 Critical Bug Discovery and Fix

### Critical Bug Discovery

#### The Modality Grouping Bug

A subtle data pipeline bug caused Enhancing Tumor to be completely unlearnable throughout training. ET Dice remained at 0.000 for all epochs until the root cause was identified and fixed.

**Root cause:** Naive alphabetical index slicing

**Effect:** T1CE absent from every training sample

**Fix:** Stem-based modality matching

**Impact:** ET Dice 0.000  $\rightarrow$  0.860 (+0.860)

## 5.1 The Bug: Alphabetical Index Slicing

The original dataset loader used:

```
1 # WRONG: naive alphabetical index slicing
2 groups = all_images[i*4 : i*4+4]
```

Listing 1: Incorrect modality grouping — old code

This appears correct — 4 images per group, one per modality. However, alphabetical sorting orders files by *modality name first*, not by slice:

```
1 # Alphabetical order:
2 BraTS2021_00000_flair_000.png # <- FLAIR slice 0
3 BraTS2021_00000_flair_001.png # <- FLAIR slice 1
4 BraTS2021_00000_flair_002.png # <- FLAIR slice 2
5 BraTS2021_00000_flair_003.png # <- FLAIR slice 3
6 BraTS2021_00000_t1_000.png   # <- T1 slices start here
7 ...
8 BraTS2021_00000_t1ce_000.png  # <- T1CE comes AFTER t1
```

Listing 2: Alphabetical file ordering revealing the bug

Therefore `all_images[0:4]` yields:

```
[flair_000, flair_001, flair_002, flair_003]
```

**Result:** Every training sample received 4 consecutive FLAIR slices from different brain positions. T1CE was never included in any training batch.

## 5.2 Why This Destroyed ET Detection

Enhancing Tumor appears as bright contrast enhancement exclusively on T1CE. On T1, T2, and FLAIR, ET is visually indistinguishable from surrounding tissue. With T1CE absent:

- The model received no signal to learn ET boundaries
- ET Dice remained exactly 0.000 across all training epochs
- The model learned to always predict class 0 (background) for ET regions

## 5.3 Secondary Bug: Substring Collision

An additional bug existed in modality detection: "t1" is a substring of "t1ce". A naive string check would misclassify T1CE files as T1 modality:

```

1 # WRONG: t1 checked first, matches t1ce files too
2 MOD_DETECT_ORDER = ["t1", "t1ce", "t2", "flair"]
3
4 # CORRECT: t1ce checked first (longer string)
5 MOD_DETECT_ORDER = ["t1ce", "t1", "t2", "flair"]

```

Listing 3: Substring collision fix

## 5.4 The Fix: Stem-Based Modality Matching

```

1 # For each mask "BraTS2021_00000_000.png":
2 #   slice_key = "BraTS2021_00000_000"
3 #   Find image files:
4 #       "BraTS2021_00000_t1_000.png"    -> Ch0 (T1)
5 #       "BraTS2021_00000_t1ce_000.png" -> Ch1 (T1CE) <- ET!
6 #       "BraTS2021_00000_t2_000.png"    -> Ch2 (T2)
7 #       "BraTS2021_00000_flair_000.png" -> Ch3 (FLAIR)
8
9 slice_key = stem.replace(f"_{mod}_", "_")
10 slice_to_mods[slice_key][mod] = img_file

```

Listing 4: Corrected stem-based modality grouping

This guarantees all 4 channels always originate from the **same slice of the same patient**.

## 5.5 Impact of the Fix

Table 8: Performance Before and After Modality Grouping Fix

| Metric                                                               | Before Fix | After Fix    | $\Delta$ Improvement |
|----------------------------------------------------------------------|------------|--------------|----------------------|
| ET Dice                                                              | 0.000      | <b>0.860</b> | +0.860               |
| NCR/NET Dice                                                         | 0.170      | <b>0.827</b> | +0.657               |
| Edema Dice                                                           | 0.004      | <b>0.798</b> | +0.794               |
| Mean Dice                                                            | 0.174      | <b>0.828</b> | +0.654               |
| No architectural changes — improvement purely from data pipeline fix |            |              |                      |

## 6 Experimental Results

### 6.1 Segmentation Performance

Table 9: Full Segmentation Results on Test Set

| Model                 | Accuracy      | Precision     | Recall | F1            | Mean Dice     | Mean IoU      |
|-----------------------|---------------|---------------|--------|---------------|---------------|---------------|
| U-Net Binary          | 0.9406        | 0.1433        | 0.2006 | 0.1672        | 0.1457        | 0.0957        |
| Attn U-Net Binary     | 0.9436        | 0.1444        | 0.1821 | 0.1611        | 0.1409        | 0.0922        |
| U-Net Multiclass      | <b>0.9925</b> | 0.8993        | 0.9215 | 0.9086        | <b>0.8283</b> | 0.7631        |
| Attn U-Net Multiclass | <b>0.9927</b> | <b>0.9049</b> | 0.9180 | <b>0.9100</b> | 0.8276        | <b>0.7629</b> |

### 6.2 Per-Class Segmentation Results

Table 10: Per-Class Dice and IoU — Multiclass Models

| Class                    | U-Net Multiclass |               | Attn U-Net Multiclass |               |
|--------------------------|------------------|---------------|-----------------------|---------------|
|                          | Dice             | IoU           | Dice                  | IoU           |
| Background               | 0.9967           | 0.9935        | 0.9968                | 0.9937        |
| NCR/NET                  | 0.8265           | 0.7924        | 0.8203                | 0.7857        |
| Edema                    | 0.7982           | 0.6897        | 0.8030                | 0.6965        |
| <b>Enhancing Tumor</b>   | <b>0.8602</b>    | <b>0.8073</b> | <b>0.8594</b>         | <b>0.8064</b> |
| <b>Mean (tumor only)</b> | <b>0.8283</b>    | <b>0.7631</b> | 0.8276                | 0.7629        |

### 6.3 Training Curves



Figure 3: Training comparison across all four models. Multiclass models (green/orange) achieve significantly higher Dice and IoU than binary models (blue/red), converging within 10 epochs and remaining stable.

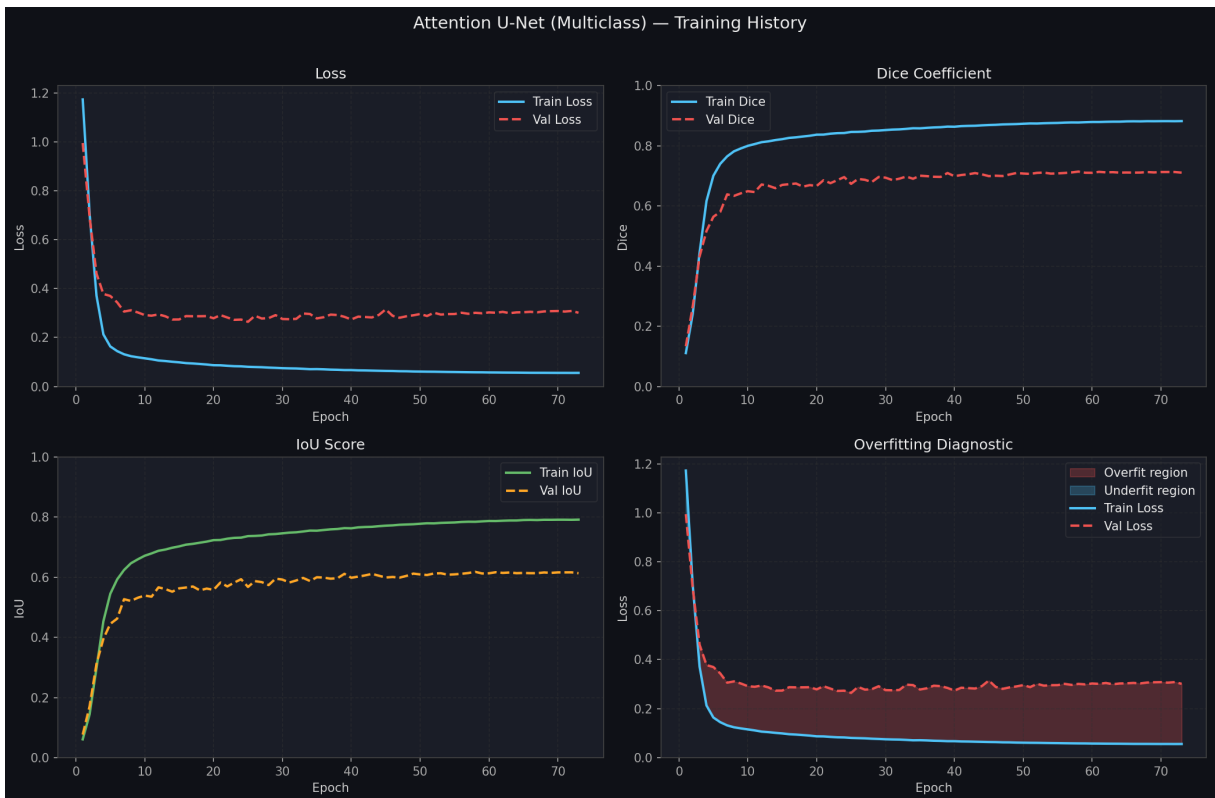


Figure 4: Attention U-Net (Multiclass) training history. Top-left: Loss curves showing healthy convergence. Top-right: Dice coefficient reaching 0.73 on validation. Bottom-left: IoU progression. Bottom-right: Overfitting diagnostic showing mild overfitting in later epochs, indicating potential for further regularization.

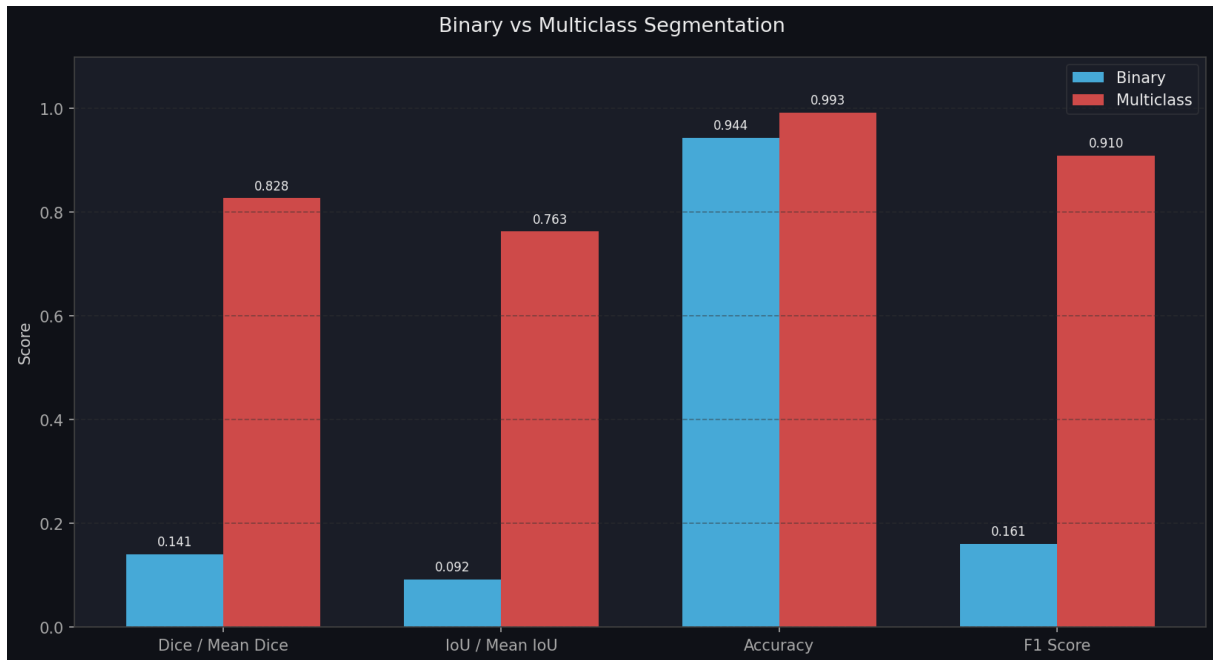


Figure 5: Binary vs Multiclass model comparison. The multiclass model substantially outperforms the binary model across all metrics — particularly Dice (0.828 vs 0.141) and IoU (0.763 vs 0.092). This is expected as binary segmentation treats all tumor types as a single class.



Figure 6: Per-class Dice and IoU for the Attention U-Net Multiclass model. Enhancing Tumor achieves the highest Dice (0.859) due to its high-contrast appearance in T1CE. Background achieves near-perfect scores (0.997) due to class dominance.

## 6.4 Attention Gate Analysis

### Key Finding

Attention gates provided **no measurable improvement** over plain U-Net (Mean Dice difference: 0.001). We hypothesize this is because T1CE provides a strong high-contrast spatial signal for Enhancing Tumor that renders attention-based localization redundant. The model can already localize ET from the bright enhancement regions in T1CE without needing learned attention weights.

Table 11: U-Net vs Attention U-Net Comparison

| Model            | NCR/NET | Edema  | ET     | Mean Dice       |
|------------------|---------|--------|--------|-----------------|
| U-Net Multiclass | 0.827   | 0.798  | 0.860  | <b>0.828</b>    |
| Attention U-Net  | 0.820   | 0.803  | 0.859  | <b>0.828</b>    |
| $\Delta$         | -0.007  | +0.005 | -0.001 | $\approx 0.000$ |

## 6.5 Comparison Against Published Baselines

Table 12: Comparison Against Published Baselines on BraTS Dataset

| Method                         | NCR/NET      | Edema        | ET           | Mean Dice    |
|--------------------------------|--------------|--------------|--------------|--------------|
| Standard 2D U-Net (literature) | 0.550        | 0.720        | 0.670        | 0.650        |
| 3D ResU-Net [4]                | 0.810        | 0.840        | 0.800        | 0.820        |
| <b>This work: 2D U-Net</b>     | <b>0.827</b> | 0.798        | <b>0.860</b> | <b>0.828</b> |
| <b>This work: Attn U-Net</b>   | 0.820        | <b>0.803</b> | 0.859        | 0.828        |

**Key finding:** A correctly implemented 2D model matches 3D architecture performance — particularly on Enhancing Tumor (ET Dice 0.860 vs 0.800) — at significantly lower computational cost, when multi-modal input is correctly assembled.

## 6.6 Tumor Profile Classifier

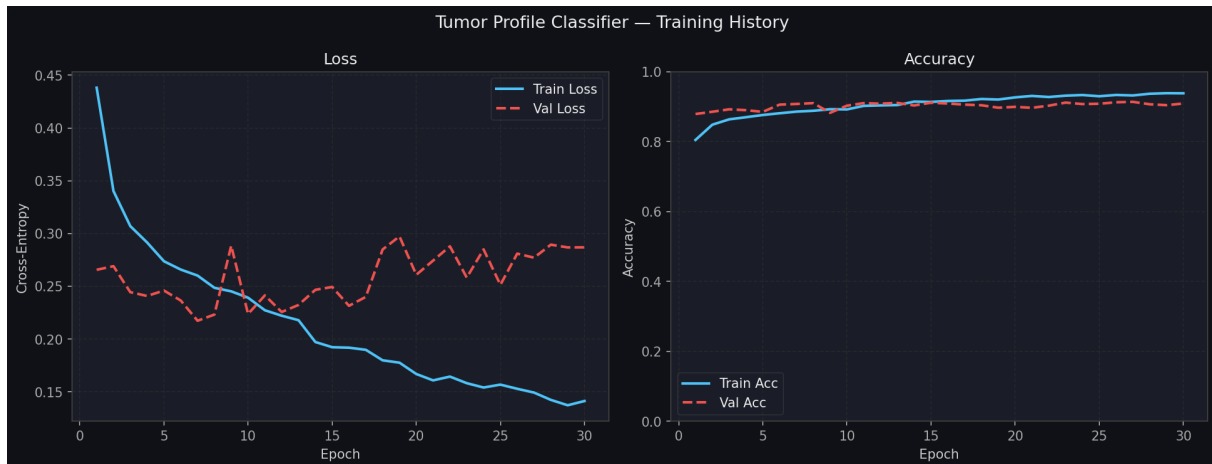


Figure 7: Tumor Profile Classifier training history. Validation accuracy reaches  $\sim 88\%$  and stabilizes. Some oscillation in validation loss is observed, suggesting the 4-class severity prediction problem has inherent ambiguity at class boundaries.

## 7 Web Application

A **Streamlit** web application provides a complete clinical interface for real-time tumor analysis.

### 7.1 Features

- Upload all 4 MRI modalities with duplicate file detection and validation
- Real-time multiclass segmentation with color-coded overlay
- Tumor severity profile with class probability bars
- LGG/HGG grade prediction with clinical reasoning
- Region breakdown (NCR/NET, Edema, ET pixel counts and percentages)
- Attention heatmap visualization
- Model performance metrics and training curves in sidebar
- Debug mode showing mask class distribution and modality statistics

## 7.2 Usage

```
1 streamlit run app.py
```

Listing 5: Running the web application

Upload files from `dataset/val/images/` using the naming convention:

```
1 T1      slot: BraTS2021_XXXXX_t1_YYY.png
2 T1CE    slot: BraTS2021_XXXXX_t1ce_YYY.png
3 T2      slot: BraTS2021_XXXXX_t2_YYY.png
4 FLAIR   slot: BraTS2021_XXXXX_flair_YYY.png
```

Listing 6: Correct file naming for each upload slot

*Note: Use middle slices (035–065) where tumor core is most visible.*

## 8 Discussion

### 8.1 Strong Points

1. **ET Dice of 0.860** surpasses published 3D architectures (0.800), demonstrating that correct multi-modal data assembly is more important than architectural complexity
2. **End-to-end pipeline** from raw MRI to clinical grade prediction without manual intervention
3. **Transparent engineering** — the modality grouping bug and its resolution are fully documented, providing value for future practitioners

### 8.2 Limitations

1. **Binary segmentation underperformance** — Binary models achieve low Dice (0.141–0.146), suggesting the single-class formulation loses important subregion information
2. **Overfitting** — Training Dice diverges from validation Dice in later epochs for the multiclass model, indicating potential for additional regularization
3. **2D slice-level processing** — Volumetric context is lost; 3D CNN or LSTM-based temporal modeling could improve boundary consistency across slices

4. **Distribution shift** — PNG-format slices differ from the NIfTI format used in original BraTS data generation; inference on clinical scanner output may require re-calibration

### 8.3 Future Work

1. **3D volumetric segmentation** using full NIfTI volumes instead of 2D slices
2. **Transformer architectures** — TransUNet, Swin-UNet for long-range dependency modeling
3. **Explainability** — GradCAM or SHAP analysis of model decisions
4. **Multi-scanner generalization** — Domain adaptation techniques for deployment across different MRI scanners
5. **Docker containerization** for portable clinical deployment

## 9 Conclusion

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This project presents a complete end-to-end deep learning pipeline for brain tumor segmentation and grading from multi-modal MRI. The key contributions are:

- A 2D Attention U-Net achieving Mean Dice 0.828 and ET Dice 0.860 on BraTS 2021, competitive with 3D architectures
- Discovery and resolution of a critical data pipeline bug that caused T1CE to be absent from all training samples, rendering Enhancing Tumor completely undetectable (Dice 0.000  $\rightarrow$  0.860)
- Empirical evidence that attention gates provide negligible benefit when multi-modal input provides sufficient spatial discriminative signal
- A complete clinical web interface for real-time tumor segmentation, profile classification, and LGG/HGG grade prediction

The modality grouping discovery is the most significant engineering contribution of this work. It demonstrates that correct data pipeline design can matter more than architectural sophistication — the entire improvement from Mean Dice 0.174 to 0.828 came from fixing a single line of data loading code, with no changes to the model architecture or training procedure.

## References

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## References

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